

franken_markdown

A clean-room, **zero-dependency** Rust engine that turns Markdown into a gorgeous self-contained HTML page *and* a professional-typography PDF — with embedded subset fonts, real kerning, ligatures, and LaTeX-grade line breaking. This document is the visual gauntlet: if it renders beautifully, the engine works.

Inline formatting

You get **bold**, *italic*, ***bold italic***, inline `code`, and ~~strikethrough~~ — composably, so you can write `bold-italic` code or a **bold link to [the docs](#)** inside a sentence. Emphasis nests correctly: **bold with *italic* inside**, *italic with **bold** inside*, and the tricky ***bold italic*** closing run all resolve the way CommonMark says.

Links come in three flavors — an inline [link](#), a reference [link](#), and a bare autolink like <https://rust-lang.org> or even <https://example.com/path?q=1> in running text. Hard breaks work too — this line ends with two spaces and wraps to a new line.

Lists, nested and mixed

Unordered lists nest cleanly:

- Top-level bullet with a fair amount of text so the line wraps and the Knuth–Plass breaker has a real paragraph to optimize across the measure.
 - A nested bullet
 - A doubly-nested bullet
 - Back to one level of nesting
- Another top-level item

Ordered lists keep their counters and nest too:

1. First step
2. Second step
 1. A sub-step
 2. Another sub-step
3. Third step

Task lists track state:

- [x] Parse the document into an AST
- [x] Shape glyphs with kerning and ligatures
- [x] Knuth–Plass line breaking

[] Liang hyphenation

Tables

GFM pipe tables become measured-column grids with per-cell alignment:

Feature	Status	Notes
HTML output	working	all-in-one, themeable
PDF output	working	embedded fonts, kerning, KP
Syntax highlight	working	a dozen-plus languages
Zero dependencies	yes	the engine has no crates

A second, denser table to stress column measuring:

Lang	Kind	Highlighted
Rust	systems	yes
Python	scripting	yes
JSON	data	yes
TOML	config	yes

Blockquotes

Blockquotes get a soft accent bar and a tinted background, and they wrap across multiple lines just like a real preview pane would render them.

Nested quotes indent further, so threaded commentary stays readable even a couple of levels deep.

Code blocks

Fenced code blocks sit on a rounded panel with **clean-room syntax highlighting** for the languages that actually show up in technical writing.

```
fn fibonacci(n: u32) -> u64 {
    let (mut a, mut b) = (0u64, 1u64);
    for _ in 0..n {
        (a, b) = (b, a + b); // tuple-destructuring update
    }
    a // the nth Fibonacci number
}
```

```
def fibonacci(n: int) -> int:
    a, b = 0, 1
    for _ in range(n):
        a, b = b, a + b # parallel assignment
    return a

{
    "engine": "franken_markdown",
    "deps": 0,
    "outputs": ["html", "pdf"],
    "fast": true
}

# Render this document to a PDF
fmd render examples/showcase.md --to pdf --out showcase.pdf
```

That's the tour. Everything above is rendered by hand-written, dependency-free Rust — the parser, the syntax highlighter, the font subsetter, and the PDF writer are all our own code.